
Name of Hand In

Course code

Author

Max Eriksson

maxerikss@gmail.com

Lund University

Department of Physics



LUND
UNIVERSITY

October 15, 2024

1 Particle Decay and Rockets

1.1 a

Problem: The relative velocity v between two particles can be determined from the four-scalar product of their four-momenta,

$$P_1 \cdot P_2 = m_1 m_2 c^2 \gamma(v)$$

(where $\gamma(v) \equiv 1/\sqrt{1 - v^2/c^2}$). Show this formula and motivate well.

Solution: Let the first particle be in rest. Letting u_1 and u_2 be the four-velocities then

$$u_1 = \gamma(0)(c, \vec{0})$$

$$u_2 = \gamma(v)(c, \vec{v}).$$

Since $\gamma(0) = 1$ we can write the four-momentum as

$$P_1 = m_1 u_1 = m_1(c, \vec{0})$$

$$P_2 = m_2 u_2 = m_2 \gamma(v)(c, \vec{v}).$$

The calculation is therefore

$$P_1 \cdot P_2 = m_1 m_2 \gamma(v)(c, \vec{0}) \cdot (c, \vec{v}) = m_1 m_2 \gamma(v)(c^2 - \vec{0} \cdot \vec{v}) = m_1 m_2 c^2 \gamma(v)$$

□

1.2 b

Problem: Consider a particle A at rest, decaying into two particles B and C. Use suitable invariants to show that the velocity of B is determined by

$$\gamma(v_B) = \frac{m_A^2 + m_B^2 - m_C^2}{2m_A m_B}$$

Solution: Before the collision we have the four momentum

$$P_A = (E_A/c, \vec{0})$$

where $E_A = m_A c^2$ and thus

$$P_A = m_A c(1, \vec{0})$$

1.3 c

Problem: Imagine that we can fuel a rocket with an ideal fuel, all of whose mass is converted to energy, leaving only massless exhaust that is released straight backwards. A rocket, that initially is at rest, is packed with ideal fuel. The fuel makes up a fraction α of the loaded rocket's total mass. Derive the maximum velocity of the empty rocket after burning all the fuel, as a function of α . (*Answer:* $v = c \frac{1-(1-\alpha)^2}{1+(1-\alpha)^2}$. *Hint:* Consider the process as a “decay”.)

Solution: